BLACKBOX TECHNIQUE

EP = Equialence partitioning

BA = Boundary values

USERNAME

Valid inputs		Invalid Inputs	
v1:	Username = alphanumeric (A-Z, a-z, 0-9)	i1:	Empty string
v2:	Username MIN length = 8 char - MAX length = 28 char	i2:	String containing special characters (e.g., "@", "!", "#")
v3:	Username can contain Danish characters (æ, ø, å)	i3:	String containing white space (e.g., "Anna 49")
		i4:	String = null
		i5:	String length < 8 or > 28
		i6:	Contains non-danish characters (José, François)

Test values		3 Boundary Values	
v1, v2, v3:	"AnneØrå9"	1.	"Abcdefg" (7 chars - **Invalid**)
i1:	""	2.	"Abcdefgh" (8 chars - **Valid**)
i4:	null	3.	"Abcdefghijklmnopqrstuvwxyzab" (28 chars - **Valid**)
i2:	"Anne123!"	4.	"Abcdefghijklmnopqrstuvwxyzabc" (29 chars - **Invalid**)
i3:	"Anna 123"		"Abcdefg1a" (9 char - Valid)
v1:	"12345678"		"Abcdefghijklmnopqrstuvwxy1a" (27 chars- Valid)
i6:	"François"		"Abcdefghijklmno1a" (18 char valid)

Total test case values:		Edge cases:
1.	"AnneØrå9"	
2.	""	
3.	null	
4.	"Anne123!"	
5.	"Anna 123"	
6.	"Abcdefg"	
7.	"Abcdefgh"	
8.	"Abcdefghijklmnopqrstuvwxyzab"	
9.	"Abcdefghijklmnopqrstuvwxyzabc"	
10.	"12345678"	
11.	"François"	
	"Abcdefg1a" (9 char - Valid)	
	"Abcdefghijklmnopqrstuvwxy1a" (27 chars- Valid)	
	"Abcdefghijklmno1a" (18 char valid)	

BLACKBOX TECHNIQUE

EP = Equialence partitioning

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PASSWORD

Valid inputs		Invalid Inputs	
v1:	Password MIN length = 8 char - MAX length = 28 char	i1:	Empty string
v2:	Contains at least one capital letter (A-Z)	i2:	Missing special character
v3:	Contains at least one number (0-9)	i3:	String containing white space (e.g., "Anna 49")
v4:	Contains at least one special character (!, #, &, etc.)	i4:	String = null
v5:	Contains Danish letters (Æ, Ø, Å)	i5:	String length < 8 or > 28
		i6:	Missing capital letter or missing number

Test values		3 Boundary Values	
v1, v2, ,v3, v4:	"Password123!"	1.	"Abcde1!" (7 chars - Invalid)
v1, v5:	"Æøå1234!"	2.	"Abcdef1!" (8 chars - Valid)
i1:	""	3.	"Abcdefghijklmnopqrstuvwxyz1!" (28 chars - Valid)
i2:	"Password123"	4.	"Abcdefghijklmnopqrstuvwxyza1!" (29 chars - Invalid)
i3:	"Pass word1!"	5.	"Abcdefg1!" (9 char - Valid)
i4:	null	6.	"Abcdefghijklmnopqrstuvwxy1!" (27 chars- Valid)
i6:	"password!!!"	7.	"Abcdefghijklmno1!" (18 char valid)

Total test case values:		Edge cases:
1.	"Password123!"	
2.	"Æøå1234!"	
3.	""	
4.	"Password123"	
5.	"Pass word1!"	
6.	null	
7.	"password!!!"	
8.	"Abcde1!"	
9.	"Abcdef1!"	
10.	"Abcdefghijklmnopqrstuvwxyz1!"	
11.	"Abcdefghijklmnopqrstuvwxyza1!"	
12.	"Abcdefg1!"	
13.	"Abcdefghijklmnopqrstuvwxy1!"	
14.	"Abcdefghijklmno1!"	

BLACKBOX TECHNIQUE

EP = Equialence partitioning

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FIRST AND LASTNAME

Valid inputs		Invalid Inputs	
v1:	MIN length = 1 char - MAX length = 30 char	i1:	Empty string
v2:	Only letters (A-Z) including Danish (Æ,Ø,Å)	i2:	Length > 30
v3:	First letter uppercase	i3:	Contains digits
v4:	Remaining letters lowercase	i4:	Contains special characters (Mads!, M@ds)
		i5:	Contains non-danish characters (José, François)
		i6:	First letter not uppercase (mads)
		i7:	More than first letter is uppercase (MADS)

Test values		3 Boundary Values	
v1:	"Abcdefghijklmno" (middle value)	1.	0, 1, 2 characters (lower boundary)
v2:	"Åse"	2.	29, 30, 31 characters (upper boundary)
v3, v4:	"Adam"		
i1:	""		
i2:	STRING MAX LENGTH		
i3:	"Mads1"		
i4:	"Mads!"		
i5:	"Madçois"		
i6:	"mads"		
i7:	"MADS"		

Total test case values:		Edge cases:
1.	"Abcdefghijklmno"	1.
2.	"Åse"	2.
3.	"Adam"	3.
4.	""	
5.	STRING MAX LENGTH	
6.	"Mads1"	
7.	"Mads!"	
8.	"Madçois"	
9.	"mads"	
10.	"MADS"	
11.	"A"	
12.	"Ab"	
13.	"Abcdefghijklmnopqrstuvwxyzaaa"	
14.	"Abcdefghijklmnopqrstuvwxyzaaaa"	
15.	"Abcdefghijklmnopqrstuvwxyzaaaaa"	

BLACKBOX TECHNIQUE

BA = Boundary values

PHONENUMBER

Valid inputs	Invalid Inputs
v1: 1-digit prefix - "2" ("21234567")	i1: Less than 8 digits length ("1234567")
v2: 2-digit prefixes - "30" ("30123456")	i2: More than 8 digits length ("123456789")
v3: 3-digit prefixes - "342" (34212345)	i3: Not in allowed list ("10123456")
v4: 3-digit prefix ranges - 344-349 ("34612345")	i4: Correct prefix but wrong length ("3012345")
v5: 31123456 - prefix 31	i5: Non-numeric characters ("30A23456")
v6: 40123456 - prefix 40	i6: Prefix partially matching but invalid - "34" ("34312345")
v7: 53123456 - prefix 53	i7: 35012345 // outside 344-349 range
v8: 71123456 - prefix 71	i8: 35812345 // outside 356-357 range
v9: 82912345 - prefix 829	i9: 49712345 // outside 493-496 and 498-499 ranges

Test values	3 Boundary Values
v1: "21234567"	1. 7,8,9 char
v2: "30123456"	2. 34912345 // upper boundary of 344-349
v3: "34212345"	3. 35612345 // lower boundary of 356-357
v4: "34612345"	4. 49812345 // lower boundary of 498-499
v5: "31123456"	
v6: "40123456"	
v7: "53123456"	
v8: "71123456"	
v9: "82912345"	
i1: "1234567"	
i2: "123456789"	
i3: "10123456"	
i4: "3012345"	
i5: "30A23456"	
i6: "34312345"	
i7: "35012345"	
i8: "35812345"	
i9: "49712345"	

Total test case values:	Edge cases:
1. "21234567"	1.
2. "30123456"	2.
3. "34212345"	3.
4. "34612345"	
5. "31123456"	
6. "40123456"	
7. "53123456"	
8. "71123456"	
9. "82912345"	
10. "1234567"	
11. "123456789"	
12. "10123456"	
13. "3012345"	
"30A23456"	
"34312345"	
"35012345"	
"35812345"	
"49712345"	
"34912345"	
"35612345"	
"49812345"	
char 8, char 9, char 7	
""	
null	

BLACKBOX TECHNIQUE

EP = Equialence partitioning

BA = Boundary values

EMAIL

Valid inputs		Invalid Inputs	
v1:	Correct format: text@text.xx	i1:	Missing @
v2:	Text before @ contains letters/numbers	i2:	Missing text before @
v3:	Domain contains letters	i3:	Missing domain
v4:	Ends with . + 2-3 letters	i4:	Missing TLD (top level domain) (.dk, .com)
		i5:	Special characters in text before @
		i6:	Special characters in domain
		i8:	Numbers in TLD
		i9:	Special characters in TLD
		i10:	TLD missing .

Total test case values:		Edge cases:
1.	"text@domain.tld"	1.
2.	"textdomain.tld"	2.
3.	"@domain.tld"	3.
4.	"text@.tld"	
5.	"text@domain"	
6.	"text!@domain.tld"	
7.	"text@domain!.tld"	
8.	"text@domain.tld1"	
9.	"text@domain.tld!"	
10.	"text@domaintld"	
11.		
12.		
13.		
14.		
15.		
16.		
17.		
18.		
19.		

Test values		3 Boundary Values
v1:	"text@domain.tld"	Irrelevant
v2, v3. v4:	"text2@domain.tld"	
i1:	"textdomain.tld"	
i2:	"@domain.tld"	
i3:	"text@.tld"	
i4:	"text@domain"	
i5:	"text!@domain.tld"	
i6:	"text@domain!.tld"	
i7:	"text@domain1.tld"	
i8:	"text@domain.tld1"	
i9:	"text@domain.tld!"	
i10:	"text@domaintld"	

BLACKBOX TECHNIQUE

EP = Equialence partitioning

BA = Boundary values

RECEIPT NUMBER

Valid inputs		Invalid Inputs	
v1:	Receipt number = 10-15 numeric characters	i1:	Receipt number = empty
v2:	Receipt number = unique	i2:	Receipt number = 1-9 numeric characters
		i3:	Receipt number = 16-MAX numreic characters
		i4:	Receipt number = alphanumeric
		i5:	Receipt number = alphabetic
		i6:	Receipt number = not unique - duplicate

Test values		3 Boundary Values	
v1:	"123456789123" (middle value)	1.	9, 10, 11 characters
v2:	"1111111111" - "2222222222"	2.	14, 15, 16 characters
i1:	""	3.	0, 1, 2 characters
i2:	"1234" (middle value)	4.	8, 9, 10 characters
i3:	"12345678912345678912" (20 characters)	5.	15, 16, 17 characters
i4:	"12345abcde"		
i5:	"abcdefghij"		
i6:	"1111111111" - "1111111111"		

Total test case values:		Edge cases:
1.	"123456789123"	1.
2.	"1111111111" - "2222222222"	2.
3.	""	3.
4.	"1234"	
5.	"12345678912345678912"	
6.	"12345abcde"	
7.	"abcdefghij"	
8.	"1111111111" - "1111111111"	
9.	1 (1 char)	
10.	11 (2 char)	
11.	11111111 (8 char)	
12.	111111111 (9 char)	
13.	1111111111 (10 char)	
14.	11111111111 (11 char)	
15.	11111111111111 (14 char)	
16.	111111111111111 (15 char)	
17.	1111111111111111 (16 char)	
18.	11111111111111111 (17 char)	

BLACKBOX TECHNIQUE

EP = Equialence partitioning

BA = Boundary values

RECEIPT PRICE

Valid inputs		Invalid Inputs	
v1:	Price = 99.00-999.99	i1:	Price = 0
v2:	Price = only 2 digits after decimal	i2:	Price > 999.99
		i3:	Price < 99.00
		i4:	price < 0
		i5:	Price = non digits

Test values		3 Boundary Values	
v1, v2:	499.99 (middle value)	1.	98.99, 99.00, 99,01
i1:	0	2.	999.98, 999.99, 1000.00
i2:	1200.95	3.	98.98, 98.99, 99.00
i3:	49.50	4.	999.99, 1000.00, 1000.01
i4:	-5	5.	-0.99, 0, 0.01
i5:	abc		

Total test case values:		Edge cases:	
1.	499.99	1.	MAX INT
2.	0		
3.	1200.95		
4.	49.50		
5.	-5		
6.	abc		
7.	98.99		
8.	99.00		
9.	99.01		
10.	999.98		
11.	999.99		
12.	1000.00		
13.	98.98		
14.	1000.01		
15.	-0.99		
16.	0.01		

BLACKBOX TECHNIQUE

EP = Equialence partitioning

BA = Boundary values

REVIEW

Value	Partition type	Partitions	Test case values	Boundary values	Test case values	Expected Result
Title	Valid: 1	1 - 60 CHAR	30 CHAR	1, 60 CHAR	1, 2, 30, 59, 60, 61 CHAR	
	Invalid: 1	Empty	""	""	Empty	ERROR
	Invalid: 2	Null	Null	Null	Null	ERROR
	Invalid: 3	61-MAX char	80 CHAR	61 CHAR	60, 61, 62 CHAR	
Comment	Valid: 1	1 - 500 CHAR	250 CHAR	1, 500 CHAR	0, 1, 250, 499, 500, 501 CHAR	
	Invalid: 1	Empty	""	""	Empty	ERROR
	Invalid: 2	Null	Null	Null	Null	ERROR
	Invalid: 3	501-MAX char	550 CHAR	61 CHAR	500, 501, 502 CHAR	
Rating	Valid: 1	1 - 10 INT	5 INT	1, 10 INT	0, 1, 2, 5, 9, 10, 11 INT	
	Invalid: 1	0 INT	0 INT	0 INT	0 INT	
	Invalid: 2	11 - MAX	11 INT	11 INT	10, 11, 12 INT	
	Invalid: 3	MIN - 1	-1 INT	-1 INT	0, -1, -2 INT	

BLACKBOX TECHNIQUE

EP = Equialence partitioning

BA = Boundary values

CARD NUMBER

Valid inputs	**Invalid Inputs**
v1: Card number = 16 numeric characters	i1: Card number = 1-15 characters
	i2: Card number = 17-MAX characters
	i3: Card number contains letters
	i4: Card number is empty
	i5: Card number is null
	i6: Card number contains special characters
	i7: Card number contains white spaces
	i8: Card number is alphanumeric
Test values	**3 Boundary Values**
v1: "1234567812345678"	1. 15, 16, 17 characters
i1: "12345678"	2. 1, 2 characters
i2: "12345678901234500000000000"	3. 14, 15 ,16 characters
i3: "abcdefghijklmnop"	4. 16, 17, 18 characters
i4: ""	
i5: null	
i6: "111111111111111#"	
i7: "111111111111111 1"	
i8: "111111111111abcd"	

Total test case values:	**Edge cases:**
1. "1234567812345678"	1. 1234567891111111 INT
2. "12345678"	2. true
3. "12345678901234500000000000"	3. 3.14
4. "abcdefghijklmnop"	
5. ""	
6. null	
7. "111111111111111#"	
8. "111111111111111 1"	
9. "111111111111abcd"	
10. "111111111111111" (15 char)	
11. "11111111111111111" (17 char)	
12. "11111111111111" (14 char)	
13. "111111111111111111" (18 char)	
14. "1" (1 char)	
15. "11" (2 char)	

BLACKBOX TECHNIQUE

EP = Equialence partitioning

BA = Boundary values

CARD MONTH

Valid inputs		**Invalid Inputs**	
v1:	Month must be between 1–12	i1:	Month = 0
		i2:	Month = Minimum to -1
		i3:	Month = 13 - MAX

Test values		**3 Boundary Values**	
v1:	6	1.	0, 1, 2
i1:	0	2.	11, 12, 13
i2:	-5	3.	-1, 0 , 1
i3:	25	4.	-2, -1, 0
		5.	12, 13, 14

Total test case values:		**Edge cases:**	
1.	6	1.	!"#¤%&/
2.	0	2.	3.14
3.	-5	3.	true
4.	25		
5.	1		
6.	2		
7.	11		
8.	12		
9.	13		
10.	-1		
11.	-2		
12.	14		

BLACKBOX TECHNIQUE

EP = Equialence partitioning

BA = Boundary values

CARD YEAR

Valid inputs	**Invalid Inputs**
v1: Year = 2026-MAX	i1: Year < 2026
	i2: Year = 0

Test values	**3 Boundary Values**
v1: 2030	1. 2025, 2026, 2027
i1: 2020	2. -1, 0, 1
i2: 0	

Total test case values:	**Edge cases:**
1. 2030	1. -1
2. 2020	2. 3.14
3. 0	3. "year"
4. 2025	
5. 2026	
6. 2027	

BLACKBOX TECHNIQUE

EP = Equialence partitioning

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CARD CVC

Valid inputs		**Invalid Inputs**	
v1:	CVC = 3 numeric Characters	i1:	CVC number = 1-2 characters
		i2:	CVC number = 4-MAX characters
		i3:	CVC number contains letters
		i4:	CVC number is empty
		i5:	CVC number is null
		i6:	CVC number contains special characters
		i7:	CVCnumber contains white spaces
		i8:	CVC number is alphanumeric
Test values		**3 Boundary Values**	
v1:	"123"	1.	2, 3, 4 characters
i1:	"12"	2.	1, 2 characters
i2:	"1234678"	3.	1, 2, 3 characters
i3:	"12H"	4.	3, 4, 5 characters
i4	""		
i5:	null		
i6:	"12?"		
i7:	"12 3"		
i8:	"1b3		

Total test case values:		**Edge cases:**	
1.	"123"	1.	123 as INT
2.	"12"	2.	true
3.	"1234678"	3.	3.14
4.	"12H"		
5.	""		
6.	null		
7.	"12?"		
8.	"12 3"		
9.	"1b3		
10.	"1111" (4 characters)		
11.	"11111" (5 characters)		